Aequa 0.3: what the code guarantees, and why

Certificate note accompanying the library

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Abstract

This note states, for every number printed by `aequa` 0.3, the result that justifies it, and proves the results that are new in this version. References of the form [I, Thm 6.1] are to the working notes *Certified inertial flow of convex fluids* (September 28, 2026); the numbering below (Lemma 2.3, Lemma 3.1, ...) is the one quoted in the source code. Each printed number carries one of three labels: **[exact]** (holds for every value of the network weights), **[quadrature]** (a converged deterministic quadrature, not an interval enclosure) and **[sampled]** (constants computed from sampled network values with a safety margin, not interval-verified). Section 7 lists what is not yet guaranteed.

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1 Setting

$\Omega \subset \mathbb{R}^2$ is bounded; its boundary consists of an outer loop and hole loops made of segments, circular arcs and smooth curves (*pieces*), with Dirichlet data u_b on every piece and zero net flux through every closed boundary component. The law is given by a potential F of the shear rate:

$$\Phi(A) = F(\sqrt{2}|A|), \qquad \dot\gamma(w) = \sqrt{2}\,|\varepsilon(w)|, \qquad A \in \mathrm{Sym}_0.$$

F' is the flow curve $\tau(\dot\gamma)$. The problem is [I, (P)] with $f = 0$: $-\operatorname{div} S + \nabla p + \hat\rho(u\cdot\nabla)u = 0$, $\operatorname{div} u = 0$, $S \in \partial\Phi(\varepsilon(u))$, $u = u_b$.

The network pair. The library returns

$$v = \operatorname{curl}\psi, \qquad \psi = \psi_{\mathrm{lift}} + \phi^2 N + \sum_k \psi_{c,k},$$

$$m = \operatorname{dev}\big(\operatorname{Airy}\chi + \hat\rho\, v\otimes v\big), \qquad \chi = \text{polynomial} + \text{networks} + \text{Michell terms} + \sum_k \chi_{c,k},$$

with Airy $\chi = \begin{pmatrix} \chi_{yy} & -\chi_{xy} \\ -\chi_{xy} & \chi_{xx} \end{pmatrix}$, ϕ a distance function vanishing on $\partial\Omega$, and corner terms at every re-entrant vertex V_k of opening $\omega_k > \pi$:

$$\psi_{c,k} = U\rho_k\,\zeta(r)\,(r/\rho_k)^{\lambda_k}(1-t^2)^2 g_k(t), \qquad \chi_{c,k} = \Sigma\rho_k^2\,\zeta(r)\,(r/\rho_k)^{\beta_k} h_k(t), \qquad t = \tfrac{2\theta}{\omega_k} - 1,$$

in local polar coordinates (r,θ), $0 < \theta < \omega_k$; ζ is a smooth cutoff, g_k, h_k are small networks of one variable, U, Σ are scales and $\lambda_k = \lambda_{\text{floor}} + \text{softplus}(\cdot)$, $\beta_k = \beta_{\text{floor}} + \text{softplus}(\cdot)$ are learned.

Gap and errors. With $a = \varepsilon(v)$,

$$\gamma = \Phi(a) + \Phi^*(m) - m : a = F(\dot\gamma(v)) + F^*\big(|m|/\sqrt{2}\big) - m : a, \qquad G = \int_\Omega \gamma\,dx,$$

and, for a solution (u^*, S^*), $E = \int D_\Phi(\varepsilon(v), \varepsilon(u^*); S^*)$, $E^* = \int D_{\Phi^*}(m, S^*; \varepsilon(u^*))$ as in [I, Def. 1.5]. The identity $\Phi^*(M) = F^*(|M|/\sqrt{2})$ is Lemma 2.2.

2 Laws

Assumption 2.1. $F : [0,\infty) \to [0,\infty)$ is convex with $F(0) = 0$, and $F = \tau_y g + F_K(g)$ with $\tau_y \geq 0$, $F_K \in C^1[0,\infty)$, $F_K'(0) = 0$, F_K' locally absolutely continuous on $(0,\infty)$ and $\int_0^1 F_K'(s)s^{-1}ds < \infty$.

All laws of the library satisfy Assumption 2.1: Newtonian, Herschel–Bulkley (power law, Bingham), Carreau, Carreau–Yasuda, Casson, Cross with $0 < m \leq 1$, and Williamson (Cross with $m = 1$).

Lemma 2.2 (radial potentials)**.** *Under Assumption 2.1,* Φ *is convex on* Sym_0 *and* $\Phi^*(M) = F^*(|M|/\sqrt{2})$ *with* $F^*(t) = \sup_{g\geq 0}(tg - F(g))$*. Consequently* $\gamma \geq 0$ *pointwise.*

Proof. $A \mapsto \sqrt{2}|A|$ is convex and F is convex and nondecreasing ($F' \geq F'(0^+) = \tau_y \geq 0$), so Φ is convex. For fixed $|A| = \varrho$, $\sup M : A = |M|\varrho$ (attained for $A \parallel M$), hence $\Phi^*(M) = \sup_{\varrho\geq 0}(|M|\varrho - F(\sqrt{2}\varrho)) = \sup_{g\geq 0}(g|M|/\sqrt{2} - F(g))$. The last claim is Fenchel–Young. □

Lemma 2.3 (guaranteed upper bound for implicit conjugates)**.** *Let* F *be convex and* $0 \leq g_{lo} \leq g_{hi}$ *with* $F'(g_{lo}) \leq t \leq F'(g_{hi})$ *(one-sided derivatives). Then*

$$F^*(t) \leq t\,g_{lo} - F(g_{lo}) + \big(t - F'(g_{lo})\big)(g_{hi} - g_{lo}).$$

Proof. $h(g) = tg - F(g)$ is concave with $h'(g) = t - F'(g)$, which is ≥ 0 for $g \leq g_{lo}$ and ≤ 0 for $g \geq g_{hi}$; so $F^*(t) = \max_{[g_{lo},g_{hi}]} h$. On that interval convexity gives $F(g) \geq F(g_{lo}) + F'(g_{lo})(g - g_{lo})$, hence $h(g) \leq tg_{lo} - F(g_{lo}) + (t - F'(g_{lo}))(g - g_{lo})$, and $g - g_{lo} \leq g_{hi} - g_{lo}$. □

Use in the code. For laws without a closed-form conjugate (Carreau, Carreau–Yasuda, Cross, Williamson) the code brackets the maximiser by doubling and 110 bisection steps and returns the right side of Lemma 2.3; for $t \geq \tau_{\sup} := \sup F'$ it returns $+\infty$. Since every theorem below is monotone in G, an over-estimated gap keeps every bound valid. F itself is closed-form except for Carreau–Yasuda and Cross, where $F(g) = \int_0^g F'$ is evaluated with a composite geometric 12-point Gauss rule on $(2^{-60}g, g]$; the omitted panel is nonnegative and below $F'(2^{-60}g)\,2^{-60}g$, and the Gauss error is at rounding level. This component is labelled [quadrature]. The check against brute-force maximisation and the Fenchel–Young equality $F(g) + F^*(F'(g)) = gF'(g)$ (defect $\sim 10^{-16}$ relative) are in the test suite.

Lemma 2.4 (unconditional floor)**.** *Under Assumption 2.1 let* $\nu_* = \inf_{g>0} \min\big(F_K''(g), F_K'(g)/g\big)$*. Then for all* $A, B \in \mathrm{Sym}_0$ *and* $S \in \partial\Phi(B)$*,* $D_\Phi(A, B; S) \geq \nu_*|A - B|^2$*. If* F_K' *is concave,* $\nu_* = \inf F_K''$*, which is the floor of [I, Prop. 4.1(i)].*

Proof. As in [I, Thm 3.4, Step 1], $D_\Phi \geq D_{\Phi_K}$ with $\Phi_K = F_K(\sqrt{2}|\cdot|)$. With $\varphi(\varrho) = F_K(\sqrt{2}\varrho)$, $\varphi'' = 2F_K''$ and $\varphi'/\varrho = 2F_K'(\dot\gamma)/\dot\gamma$, so [I, Lemma 3.3] gives $D^2\Phi_K \geq 2\nu_* I$ wherever defined. The Taylor argument of [I, Thm 3.4, Step 3] (which uses only Assumption 2.1) gives $D_{\Phi_K}(A,B) = \int_0^1 (1-t) D^2\Phi_K(x_t)[A-B, A-B]\,dt \geq \nu_* |A-B|^2$. If F_K' is concave with $F_K'(0) = 0$, then $F_K''(g) \leq F_K'(g)/g$. □

Lemma 2.5 (concave flow curves)**.** *(a) Carreau–Yasuda, $\tau = \eta_\infty\dot\gamma + (\eta_0 - \eta_\infty)\lambda^{-1}h(\lambda\dot\gamma)$ with $h(x) = x(1+x^a)^\mu$, $\mu = (n-1)/a$, $a > 0$: for $0 < n \leq 1$, $h'' \leq 0$ on $[0,\infty)$; so the flow curve is concave, with $\inf\tau' = \eta_\infty$ for $n < 1$. For $n > 1$, $\tau' \geq \eta_0$ and $\tau/\dot\gamma \geq \eta_0$.*
(b) Cross, $\tau = \eta_\infty\dot\gamma + (\eta_0 - \eta_\infty)\lambda^{-1}k(\lambda\dot\gamma)$ with $k(x) = x/(1+x^m)$: for $0 < m \leq 1$, $k'' \leq 0$ and $\inf\tau' = \eta_\infty$. For $m > 1$ and $\eta_\infty = 0$, $\tau' < 0$ for large $\dot\gamma$ (F is not convex).

Proof. (a) $h' = (1+x^a)^{\mu-1}(1+nx^a)$ and

$$h'' = a\,x^{a-1}(1+x^a)^{\mu-2}\big[(n-1)(1+\tfrac{1}{a}) + \tfrac{n(n-1)}{a}x^a\big],$$

both brackets are ≤ 0 for $n \leq 1$. For $a = 2$ this is [I, Lemma 3.7]. As $x \to \infty$, $h'(x) \to 0$ for $n < 1$. For $n > 1$: $1 + nx^a \geq 1 + x^a$, so $h' \geq (1+x^a)^\mu \geq 1$, and $h(x)/x = (1+x^a)^\mu \geq 1$. (b) $k' = (1 + (1-m)x^m)/(1+x^m)^2$ and $k'' = mx^{m-1}\big[-(1+m) - (1-m)x^m\big]/(1+x^m)^3 \leq 0$ for $m \leq 1$; $k' \to 0$. For $m > 1$, $k' < 0$ once $x^m > 1/(m-1)$. □

The code rejects Cross laws with $m > 1$ (this also excludes some convex cases with large η_∞; it is the safe side).

Lemma 2.6 (thickening weight in the shear-rate normalisation)**.** *For Herschel–Bulkley with $1 < n \leq 2$, $D_\Phi(\varepsilon(v),\varepsilon(u);S) \geq \nu_{\rm th}|\varepsilon(v) - \varepsilon(u)|^2$ with $\nu_{\rm th} = 2Kc_q\,\dot\gamma(v)^{n-1}$, $c_q \geq 1/24$, a weight that depends on the network only.*

Proof. $\Phi_K = K(\sqrt{2}|A|)^r/r$ with $r = n+1 \in (2,3]$. [I, Lemma 3.8] gives $D_{\Phi_K}(A,B) \geq K2^{r/2}c_q|A|^{r-2}|A-B|^2$ with $A = \varepsilon(v)$, and $2^{r/2}|A|^{n-1} = 2^{(n+1)/2}(\dot\gamma/\sqrt{2})^{n-1} = 2\dot\gamma^{n-1}$. □

Tube weight. For concave flow curves, [I, Cor. 3.5] gives $D_\Phi \geq F_K''(\sqrt{2}(|a|+t))\,t^2$ with $t = |\varepsilon(e)|$. Where $t \leq s := s_0U/L + \lambda|a|$ this is $\geq \nu_s t^2$ with $\nu_s = F_K''\big(\sqrt{2}((1+\lambda)|a| + s_0U/L)\big)$; the code uses $\max(\nu_s, \nu_*)$.

law	concave	floor ν_*	growth p
Newtonian μ	yes	μ	2
Herschel–Bulkley, $n \leq 1$	yes	K if $n = 1$, else 0	$n+1$
Herschel–Bulkley, $1 < n \leq 2$	no	0 (weight of Lemma 2.6)	$n+1$
Carreau(–Yasuda), $n \leq 1$	yes	η_∞	2 if $\eta_\infty > 0$, else $n+1$
Carreau(–Yasuda), $n > 1$	no	η_0	$n+1$
Casson	yes	K	2
Cross, Williamson, $m \leq 1$	yes	η_∞	2 if $\eta_\infty > 0$; else $2-m$

Table 1: Law data used by the code. p is the growth of F at infinity and $p' = p/(p-1)$. Only Cross with $\eta_\infty = 0$, $m = 1$ has $p = 1$, $p' = \infty$ and a saturating flow curve, $\tau_{\rm sup} = \eta_0/\lambda$; all other laws have $\tau_{\rm sup} = \infty$.

3 Admissibility, and corner exponents

Lemma 3.1 (exact admissibility; exponent floors)**.** *For every value of the weights:*

(a) $\operatorname{div} v = 0$ *in* Ω *and* $v = u_b^{\rm imp}$ *on every piece, where* $u_b^{\rm imp}$ *are the imposed data of Section 4 (*$|u_b^{\rm imp} - u_b| \leq \delta_{\rm data}$*, reported);*

(b) $\operatorname{Airy}\chi$ *is symmetric and divergence free in* $\mathcal{D}'(\Omega)$*; this includes the Michell terms of [I, Prop. 21.2], whose stresses are single valued;*

(c) if $\lambda_k > \lambda_{\min} = \max(1, 2-2/p)$ *and* $\beta_k > \beta_{\min} = 2-2/p'$ *(*$\beta_{\min} = 2$ *when* $\tau_{\sup} < \infty$*), then near* V_k*:* $\nabla v \in L^q$ *for some* $q > \max(2,p)$*,* $F(\dot\gamma(v)) \in L^1$*,* $F^*(|\operatorname{Airy}\chi_c|/\sqrt{2}) \in L^1$*, and the gap density satisfies* $\gamma \lesssim r^{-\kappa_k}$ *with*

$$\kappa_k = \max\big(0,\ (2-\lambda_k)p,\ (2-\beta_k)p',\ 4-\lambda_k-\beta_k\big) < 2.$$

The code sets $\lambda_k = \lambda_{\text{floor}} + \operatorname{softplus}(\cdot)$ *with* $\lambda_{\text{floor}} = \lambda_{\min} + 0.02$ *(and the same for* β*), so the hypotheses of (c) hold for every weight. Label: [exact].*

Proof. (a) $\operatorname{div}\operatorname{curl} = 0$. The lift has the data (Section 4); $\phi^2 N$ and its gradient vanish on $\partial\Omega$. For the corner term, $(1-t^2)^2$ and its θ-derivative $\propto t(1-t^2)$ vanish at $t=\pm1$, i.e. on both walls of the corner, and the radial derivative carries the factor $(1-t^2)^2$; so $\psi_{c,k} = \nabla\psi_{c,k} = 0$ on the walls. (b) $\operatorname{div}\operatorname{Airy}\chi = (\chi_{yyx} - \chi_{xyy},\ -\chi_{xyx} + \chi_{xxy}) = 0$ for single-valued χ with $\nabla^2\chi \in L^1_{loc}(\Omega)$; the vertices are not points of Ω. For the Michell terms see [I, Prop. 21.2]: their Hessians are single valued away from the hole centres, which lie outside Ω. (c) Inside the cutoff, $\nabla^j\psi_{c,k} = O(r^{\lambda_k - j})$, so $\nabla v = O(r^{\lambda_k-2}) \in L^q$ for $q < 2/(2-\lambda_k)$ (any q if $\lambda_k \ge 2$). $\lambda_k > 1$ gives some $q > 2$ and $\lambda_k > 2-2/p$ gives some $q > p$. $F(\dot\gamma) \le c(1+\dot\gamma^p) = O(1+r^{(\lambda_k-2)p})$ is integrable in two dimensions iff $(\lambda_k-2)p > -2$. Likewise $|\operatorname{Airy}\chi_{c,k}| = O(r^{\beta_k-2})$ and $F^*(t) \le c(1+t^{p'})$; if $\tau_{\sup} < \infty$, $\beta_k > 2$ makes the corner stress tend to 0, so it stays below $\tau_{\sup}$ near V_k. The remaining parts of v and m are bounded near V_k (the stress networks take only smooth inputs, so the defect of [I, Remark 20.6(ii)] cannot occur), and $|m:a| \le |m||a|$ gives the last exponent. Finally $(2-\lambda_k)p < 2$, $(2-\beta_k)p' < 2$ and $4-\lambda_k-\beta_k < 4-(2-2/p)-(2-2/p') = 2$. □

Lemma 3.2 (singular networks in the theorems of [I])**.** *[I] states Theorem 2.1 for* $v \in W^{1,\infty}$*. Its proof uses only:* $m, S^* \in L^{p'}$*,* $\varepsilon(v) \in L^p$*, and, when* $\hat\rho > 0$*,* $\int e\cdot\varepsilon(v)e$ *finite and (EI) tested with* $\ell = v$*. In two dimensions these hold when* $\nabla v \in L^q$ *with*

$$q \ge p \ \ (\hat\rho = 0), \qquad q \ge \max(p,\ 2,\ t^*) \ \ (\hat\rho > 0), \qquad t^* = \frac{p^*}{p^*-2},\ p^* = \frac{2p}{2-p}\ (p<2).$$

For corner terms this is $\lambda_k > \lambda_{\min}$ *when* $\hat\rho = 0$*, and* $\lambda_k > \lambda^{\text{in}}_{\min} = \max(\lambda_{\min}, 4/p-2)$ *when* $\hat\rho > 0$*; the two differ only for* $1 < p < 4/3$*. At a vertex with a data jump (*$p<2$*, Lemma 4.2),* $\nabla v = O(r^{-1})$ *and only* $q < 2$ *is available.*

Proof. For $p<2$, $e \in W^{1,p}_0 \subset L^{p^*}$, so $e\cdot\varepsilon(v)e \in L^1$ when $2/p^* + 1/q \le 1$, i.e. $q \ge t^*$; for $p \ge 2$ every L^s, $s<\infty$, is available and $q > 2$ suffices. In (EI), $\int S^* : \varepsilon(u^*-\ell)$ needs $\varepsilon(\ell) \in L^p$ and $\int(u^*\otimes u^*) : \nabla\ell$ needs $\nabla\ell \in L^{(p^*/2)'} = L^{t^*}$; (EI) extends from $W^{1,\infty}$ to such ℓ by density, since every term is continuous in that topology [I, proof of Thm 16.3]. The integration by parts in [I, Thm 2.1, Step 2] needs $v \in L^\infty$, $\nabla v \in L^q$, $e \in L^{q'}$. With $q < 2/(2-\lambda_k)$: $2/(2-\lambda_k) > t^* \iff \lambda_k > 2-2/t^* = 4/p^* = 4/p-2$. □

The code uses $\lambda^{\text{in}}_{\min}$ as the floor whenever $\hat\rho > 0$, and reports *not certified* for $\hat\rho > 0$ when the data jump at a vertex: the θ enclosure of [I, Thm 20.2] has corner cells only at re-entrant corners, and $|\varepsilon(v)| \sim 1/r$ at a jump.

4 Boundary data

The lift is the transfinite Hermite construction of [I, Prop. 21.1], with three additions.

Lemma 4.1 (closure)**.** *The user data are represented exactly on segments and by* C^2 *splines on curved pieces. On each closed boundary component the net flux of the represented data is computed (Simpson's rule, 8001 points per piece) and removed by a uniform normal velocity on*

that component. The resulting imposed data u_b^{imp} have zero flux through every closed component, so the stream-function values on each component are single valued and the lift of [I, Prop. 21.1] applies; δ_{data} (Remark 4.4) includes the correction.

Proof. A uniform normal velocity c on a closed component Γ adds the flux $c|\Gamma|$; choose c to cancel the computed flux. The stream function is the integral of $u_b^{\mathrm{imp}} \cdot n$ along Γ, which returns to its initial value after one turn exactly when the flux vanishes. □

Lemma 4.2 (vertex data)**.** *Let the data jump at a boundary vertex V: $u_b^-(V) \neq u_b^+(V)$. Then every admissible velocity has $\int_{\Omega \cap B_\delta(V)} F(\dot\gamma) = \infty$ if $p \geq 2$, while for $p < 2$ the lift satisfies $\nabla v = O(r^{-1})$ and $F(\dot\gamma(v)) = O(r^{-p}) \in L^1$.*

Proof. For $p > 2$, $W^{1,p} \subset C^0(\overline{\Omega})$ excludes a jump. For $p = 2$, the trace space of H^1 near V is $H^{1/2}$ of the boundary, and the Gagliardo seminorm of a step diverges logarithmically; since $F(\dot\gamma) \geq c\dot\gamma^2 - c'$, the dissipation is infinite. For $p < 2$: near V the blending weights of the lift are homogeneous of degree 0 in local polar coordinates, and the stream-function values on the two pieces agree at V, so $\psi_{\mathrm{lift}} - \psi(V) = O(r)$ with an angular profile; hence $\nabla v = O(r^{-1})$, and $\int_0^\delta r^{-p} r\, dr < \infty$. □

The code raises an error for a jump when $p \geq 2$, and for $p < 2$ grades the quadrature at the vertex with $\kappa = \max(p, 1)$ (Section 5).

Remark 4.3 (no-slip vertex blending)**.** At a vertex between two no-slip pieces the stream function equals a constant c with zero gradient on both walls. The lift is replaced by $c + (1 - \mathrm{cut})(\psi_{\mathrm{lift}} - c)$, with $\mathrm{cut} = 1 - s^4(35 - 84s + 70s^2 - 20s^3)$, $s = \mathrm{clamp}(2r/r_b - 1, 0, 1)$. Since $\psi_{\mathrm{lift}} - c$ and its gradient vanish on both walls, so do those of the product: the data stay exact. For $r \leq r_b/2$ the lift is exactly c, which removes the cancellation of $O(u/r^2)$ terms in its second derivatives near the vertex (tested: gradient and Hessian are exactly zero there).

Remark 4.4 (data mismatch)**.** The certificate concerns the problem with data u_b^{imp}. The code reports $\delta_{\mathrm{data}} = \max(\|u_b^{\mathrm{imp}} - u_b\|_\infty$ over the spline representation, the closure correction), typically 10^{-12}. No theorem in this note converts this into a bound for the problem with data u_b.

5 Quadrature

Proposition 5.1 (the rule)**.** *The rule of* `quadrature.py` *has positive weights and integrates over a region whose boundary equals $\partial\Omega$ exactly for polygons and circular arcs. It is assembled from: graded sectors at every straight vertex (polar Gauss panels, geometric towards the vertex down to $t_{\min} = 10^{-8}\rho_V$ at singular vertices, $10^{-6}\rho_V$ elsewhere, uniform outside); polar bands between every circle and its chord polygon; and tensor Gauss cells on the remainder, with cut cells clipped to the remaining polygon and triangulated (Dunavant degree-5 rule, positive weights).*

Proof. Each part is a positive rule on a region described exactly (sectors and bands by polar maps with Jacobian $r > 0$; remainder cells by exact polygon clipping). The parts have disjoint interiors by construction (overlaps between bands are checked and refused), and their union is Ω (tested: the area is exact to 10^{-10} relative on five geometries). □

Lemma 5.2 (sector tails)**.** *On a sector $\{0 < r < r_1,\ 0 < \theta < \omega\}$ let $f = \sum_i r^{-\kappa_i} g_i(\theta)$ with $g_i \geq 0$ and $\kappa_i \leq \kappa < 2$, and let I_1 be the integral over the innermost panel $t_{\min} < r < q t_{\min}$. Then*

$$\int_{\{r < t_{\min}\}} f \leq \frac{I_1}{q^{2-\kappa} - 1}.$$

Proof. For one term, the integrals over the panels $(q^{-j} t_{\min}, q^{-j+1} t_{\min})$ form a geometric sequence with ratio $q^{-(2-\kappa_i)}$, so the tail equals $I_1^{(i)}/(q^{2-\kappa_i} - 1) \leq I_1^{(i)}/(q^{2-\kappa} - 1)$. Sum over i. □

The leading part of the gap density at an enriched corner or a jump vertex is exactly homogeneous inside the cutoff (Lemma 3.1(c), Lemma 4.2). The code uses $\kappa = \max(\kappa_{\mathrm{fit}}, \kappa_{\mathrm{theory}})$, where κ_{fit} is read off the two innermost panels. Because the density is not exactly of the form of Lemma 5.2, and the panel integrals are Gauss approximations, the tail is labelled [quadrature].

Remark 5.3 (curved pieces)**.** Non-circular curves are represented by polylines of 4000 segments; the code reports $e_{\mathrm{geo}} = \sum \text{chord} \times \text{sag}$, which bounds the area between curve and polyline for smooth curves, and adds $e_{\mathrm{geo}} \max \gamma$ to the gap.

The value used. With the rule at spacing h (24 geometric levels) and at $2h$ (12 levels),

$$G_{\mathrm{up}} = G_h + |G_h - G_{2h}| + \sum \text{tails} + e_{\mathrm{geo}} \max \gamma.$$

The code stops if γ is not finite at some point. G_{up} is a converged quadrature value, labelled [quadrature]; it is not an interval enclosure of G.

6 Statements printed by the code

Proposition 6.1 (energy enclosure, $\hat{\rho} = 0$)**.** *Let $\hat{\rho} = 0$, let (v, m) be admissible with $G < \infty$, and let $J(w) = \int F(\dot{\gamma}(w))$ over divergence-free w with $w - v \in W_0^{1,p}$ ($W_0^{1,1}$ if $p = 1$). Then $J(v) - G \le \inf_w J(w) \le J(v)$. No existence of a minimiser is needed.*

Proof. $m \in L^{p'}$ (since $F^*(|m|/\sqrt{2}) \in L^1$ and F^* grows like $t^{p'}$; $m \in L^\infty$ if $p = 1$) and $m = \operatorname{dev} \operatorname{Airy} \chi$ is equilibrated, so $\int m : \varepsilon(w - v) = 0$. Fenchel–Young gives $J(w) \ge \int m : \varepsilon(w) - \int \Phi^*(m) = \int m : \varepsilon(v) - \int \Phi^*(m) = J(v) - G$. □

Theorem 6.2 (what is printed, by regime)**.** *Let $\nu_0 \ge 0$ be a weight with $E \ge \int \nu_0 |\varepsilon(u^* - v)|^2$ for the solutions considered, and let*

$$\theta[\nu_0] = \sup_{w \in V \setminus 0} \frac{-\hat{\rho} \int w \cdot \varepsilon(v) w}{\int \nu_0 |\varepsilon(w)|^2}.$$

(i) $\hat{\rho} = 0$: $G = E + E^*$ for every solution [I, Thm 2.1] ($p > 1$; for $p = 1$, every solution in $W^{1,1}$ with bounded stress), and Proposition 6.1.

(ii) $\hat{\rho} > 0$, $p > 1$, no data jumps, $\theta[\nu_0] < 1$: $E + E^* \le G/(1 - \theta[\nu_0])$ and $\int \nu_0 |\varepsilon(u^* - v)|^2 \le G/(1 - \theta[\nu_0])$.

(iii) $1 \le \theta < 2$: existence and uniqueness near v hold [I, Thm 9.1, 17.1], but the bound needs the residual $|m - S_a|$, which 0.3 does not evaluate; no number is printed. $\theta \ge 2$: not certified.

The weights used are: ν_* (Lemma 2.4, unconditional); ν_{th} (Lemma 2.6, unconditional); ν_s (tube weight, valid for the solutions with $|\varepsilon(u^* - v)| \le s$ a.e.; for the others alternative (a) of [I, Cor. 7.4] holds).

Proof. (ii) By [I, Thm 2.1] (valid for the singular networks by Lemma 3.2), $E + E^* \le G + \theta \int \nu_0 |\varepsilon(e)|^2 \le G + \theta E \le G + \theta(E + E^*)$. This is the proof of [I, Thm 6.1] with ν_* replaced by ν_0, and of [I, Cor. 7.4] with $P = \emptyset$; it uses neither concavity nor the power-law form beyond the weight. □

θ. The code bounds $\theta[\nu_0]$ by the dual enclosure [I, Thm 20.2] with $\tilde{\nu} = \nu_* f(\rho) \le \nu_0$ and the radial Korn factor of [I, Lemma 20.1]. The line stresses used there are exactly equilibrated only if their free ends lie outside Ω; the code moves the ends a margin of $10^{-9} \times$range plus twice the largest polyline sag outside, and refuses a line whose end falls inside. The cell majorants P_K, the floor ν_* (a sampled minimum, multiplied by 0.95) and the Gram matrix D use sampled network values: label [sampled].

Power and pressure drop ($\hat\rho = 0$)

For $t \ge 0$ let $V_t = \{w \in W^{1,p} : \operatorname{div} w = 0,\ w = t\,u_b^{\mathrm{imp}} \text{ on } \partial\Omega\}$ and $J^*(t) = \inf_{V_t} J$. For a solution (u^*, S^*) with data u_b^{imp} the *power* is

$$W^* = \int_\Omega S^* : \varepsilon(u^*)\,dx.$$

Lemma 6.3 (exponent bracket)**.** *If $q_{lo} \le g\tau'(g)/\tau(g) \le q_{hi}$ for a.e. $g > 0$, then $p_{lo} := q_{lo} + 1 \le gF'(g)/F(g) \le q_{hi} + 1 =: p_{hi}$, and $F(tg) \le t^{p_{hi}} F(g)$ for $t \ge 1$, $F(tg) \le t^{p_{lo}} F(g)$ for $0 \le t \le 1$. The code uses*

law	$[q_{lo}, q_{hi}]$	$[p_{lo}, p_{hi}]$
Newtonian; power law	$\{1\}$; $\{n\}$	$\{2\}$; $\{n+1\}$
Herschel–Bulkley, $\tau_y > 0$	$[0, n]$	$[1, n+1]$
Carreau, Carreau–Yasuda	*between n and 1*	*between $n+1$ and 2*
Casson, $\tau_y > 0$	$[0, 1]$	$[1, 2]$
Cross, Williamson	$[1-m, 1]$	$[2-m, 2]$

Proof. For $0 < s \le g$, $\log\tau(g) - \log\tau(s) = \int_s^g \tau'/\tau \le q_{hi}\log(g/s)$, so $\tau(s) \ge \tau(g)(s/g)^{q_{hi}}$ and $F(g) = \int_0^g \tau \ge g\tau(g)/(q_{hi}+1)$; likewise $F(g) \le g\tau(g)/(q_{lo}+1)$. With $\psi(\sigma) = \log F(e^\sigma)$, $\psi' = gF'/F \in [p_{lo}, p_{hi}]$; integrate ψ' between $\log g$ and $\log(tg)$. Table: for Herschel–Bulkley $g\tau'/\tau = nKg^n/(\tau_y + Kg^n)$; for Carreau–Yasuda $g\tau'/\tau = 1 + g\eta'/\eta$ with $g\eta'/\eta = (n-1)\frac{x^a}{1+x^a} \cdot \frac{(\eta_0-\eta_\infty)(1+x^a)^\mu}{\eta}$, both factors in $[0,1]$; for Cross $g\eta'/\eta = -m\frac{x^m}{1+x^m} \cdot \frac{(\eta_0-\eta_\infty)/(1+x^m)}{\eta} \in [-m, 0]$; for Casson $g\tau'/\tau = \sqrt{Kg}/(\sqrt{\tau_y} + \sqrt{Kg})$. □

Theorem 6.4 (certified power)**.** *Let $\hat\rho = 0$ and let (v, m) be admissible with gap G. For every solution and every $h \in (0,1)$,*

$$\frac{J(v) - G - J((1-h)v)}{h} \le W^* \le \frac{J((1+h)v) - J(v) + G}{h}, \qquad p_{lo}\,(J(v) - G) \le W^* \le p_{hi}\,J(v),$$

where $J(tv) = \int F(t\dot\gamma(v))$ is computable from the network. For homogeneous laws ($p_{lo} = p_{hi} = p$: Newtonian, power law) $W^* = pJ^*(1)$ and the enclosure has relative width $G/(J(v) - G)$.

Proof. W^* is a subgradient of J^* at $t = 1$. For $w \in V_t$, $S^* \in \partial\Phi(\varepsilon(u^*))$ gives $J(w) \ge J(u^*) + \int S^* : \varepsilon(w - u^*)$. Since $w - tu^*$ is divergence free with zero trace and S^* is equilibrated ($\hat\rho = 0$), $\int S^* : \varepsilon(w - tu^*) = 0$, so $\int S^* : \varepsilon(w - u^*) = (t-1)W^*$. With $t = 1$ this shows $J(u^*) = J^*(1)$, and in general $J^*(t) \ge J^*(1) + (t-1)W^*$. *Bounds for J^*.* $tv \in V_t$ gives $J^*(t) \le J(tv)$; Proposition 6.1 gives $J^*(1) \ge J(v) - G$. Taking $t = 1 \pm h$ gives the first pair. *Structural pair.* $tu^* \in V_t$ and Lemma 6.3 give $J^*(1+h) \le (1+h)^{p_{hi}} J^*(1)$ and $J^*(1-h) \le (1-h)^{p_{lo}} J^*(1)$; insert into the subgradient inequality and let $h \to 0$: $p_{lo}J^*(1) \le W^* \le p_{hi}J^*(1)$, then use $J(v) - G \le J^*(1) \le J(v)$. □

Corollary 6.5 (pressure drop)**.** *$W^* = \int S^* : \varepsilon(z)$ for every divergence-free z with data u_b^{imp} (since $z - u^*$ has zero trace), so W^* is the work done by the boundary tractions; for smooth solutions $W^* = -\int_{\partial\Omega} (\sigma^* n) \cdot u_b^{\mathrm{imp}}\,ds$ with $\sigma^* = S^* - p^* I$. If the walls are at rest and the data on inflow and outflow sections are normal to them, then*

$$W^* = Q_{in}\,\Delta p, \qquad \Delta p := \langle -\sigma^*_{nn}\rangle_{in} - \langle -\sigma^*_{nn}\rangle_{out},$$

with flux-weighted averages $\langle f\rangle = \int f\,|u_b \cdot n|\,ds/Q_{in}$ and Q_{in} the inflow flux. Hence Theorem 6.4 divided by Q_{in} encloses Δp. In a developed section $\varepsilon_{nn} = 0$, so $-\sigma_{nn} = p$ and Δp is the usual drop of the mean pressure.

Proof. $(\sigma^* n) \cdot u_b = \sigma^*_{nn}(u_b \cdot n)$ on the sections (tangential data vanish), and $u_b = 0$ on walls; zero net flux gives $Q_{in} = Q_{out}$. □

In the code. `Solver.power_bounds` evaluates $J(tv)$ on both quadrature rules (with tails and the fine–coarse difference as margin), takes $J(v) - G$ with $G = G_{\mathrm{up}}$, maximises the lower and minimises the upper bound over 40 values $h \in [10^{-3}, 0.5]$, and intersects with the structural pair. Label: [quadrature]. On briefly trained channel networks the relative width is 8.6×10^{-4} (Newtonian, $G/J = 8.6 \times 10^{-4}$), 5.6×10^{-2} (Carreau) and 7.3×10^{-2} (Herschel–Bulkley, $G/J = 3.1 \times 10^{-3}$): for laws with two exponents the two-point bound behaves like $\sqrt{G/J}$. With inertia no minimum principle is available and the power is not enclosed in 0.3.

7 Evidence and limits

Tests (all pass).

- Identity $G = E + E^*$ against exact developed channel flows for eight laws, random networks: defect $\leq 2.1 \times 10^{-10}$ relative; $E, E^* \geq 0$. Concentric Couette flow with Michell terms: quadrature-limited, $< 10^{-8}$.
- Admissibility for random networks: $|\operatorname{div} v|$ and $|\operatorname{div} \sigma|$ below 10^{-9} relative on three domains (holes, re-entrant corners, curved walls); data mismatch $< 10^{-9}$; exponent floors hold with all corner parameters at -50; lid jump rejected for $p = 2$, accepted for $p = 1.5$; exact blending.
- Quadrature: areas exact to 10^{-10} on five geometries; $r^{-\kappa}$ at a re-entrant corner, $\kappa = 0.5, 1.2, 1.7$, and $r^{-1.5}$ at a lid jump, against one-dimensional polar integrals: $\leq 3.6 \times 10^{-9}$.
- Boundary data on nine benchmark geometries.
- Power: the enclosure of Theorem 6.4 contains the exact value for seven laws (random networks, exact channel flows).

Not guaranteed in 0.3.

1. No interval arithmetic: G is [quadrature], θ is [sampled]. For saturating laws, finiteness of $F^*(|m|/\sqrt{2})$ is checked at the quadrature points only.
2. No bound in the hybrid regime $1 \leq \theta < 2$ and no check for $\theta \geq 2$.
3. Power and pressure drop are certified for $\hat{\rho} = 0$ only (Theorem 6.4); pointwise pressure is not.
4. The certificate refers to the imposed data u_b^{imp}, not to u_b.
5. Inertia together with data jumps is refused.